**Code No: 117BX** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech IV Year I Semester Examinations, July/August - 2022**
**COMPUTER GRAPHICS**
**(Computer Science and Engineering)**

**Time: 3 Hours** **Max.Marks:75**

**Answer any five questions**
**All questions carry equal marks**
**- - -**

1.a) List and explain the operating characteristics of LCDs.
b) Explain a flood-fill algorithm to fill the interior of any specified area. [8+7]

2.a) Explain the mid-point circle generation algorithm.
b) Describe various approaches used to represent polygon. [8+7]

3.a) What is the significance of 4-bit region code in Cohen-Sutherland algorithm?
b) Derive the transformation matrix for reflection about y = x line. [8+7]

4.a) What is meant by homogeneous co-ordinate system. What is its significance?
b) Write a short note on shear transformations in 2D. [8+7]

5.a) Write a short note on Bezier curves.
b) Explain about 3 D composite transformations. [8+7]

6.a) Derive the matrices for rotations about three principal axis in 3-D graphics.
b) Write a short note on specular reflection. [8+7]

7.a) What are the underlying concepts of the subdivision algorithm? Explain.
b) Write a short note on BSP tree method. [8+7]

8.a) Discuss how the kinematic description are simulated in key frame system?
b) Explain various methods used by various languages to control an animation. [8+7]

**---oo0oo---**

JNTUH Used Papers